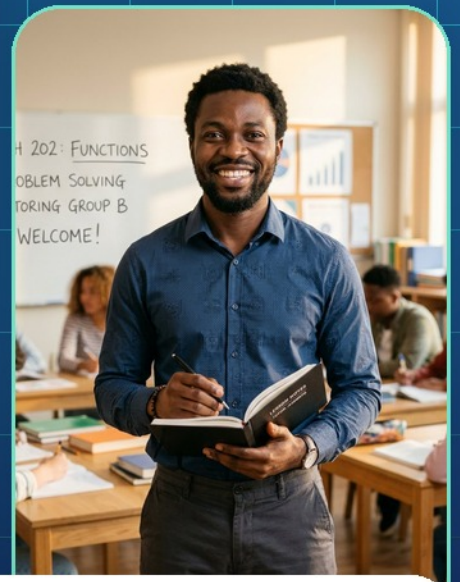
EDUMENTRY AI HACKATHON 1.0

TUTORCONNECT NIGERIA

A full-stack EdTech marketplace connecting Nigerian learners and parents with verified local and online tutors.





TEAM OF FOUR

1. **Uchekwe Uchechukwu Hephzibah**
   Team Leader & Frontend Developer
2. **Chioma Helen Ezeufonna**
   Backend Developer
3. **Uzochukwu Mary Cynthia Enyemakachukwu**
   Database Developer
4. **Bodman Ebuka Precous**
   Deployment Specialist

One story. Four specialists. Five speaking turns.

Planned duration: approximately 15 minutes • the team leader opens and returns to close.

01 — OPEN

Uchekwe Uchechukwu Hephzibah

Introduction + frontend

Slides 1-8

≈ 5 min

02 — BACKEND

Chioma Helen Ezeufonna

API + authentication

Slides 9-12

≈ 3 min

03 — DATABASE

Uzochukwu Mary Cynthia Enyemakachukwu

Schema + data flow

Slides 13-16

≈ 3 min

04 — DEPLOY

Bodman Ebuka Precous

Hosting + reliability

Slides 17-20

≈ 2.5 min

05 — RETURN

Uchekwe Uchechukwu Hephzibah

Revenue + conclusion

Slides 21-22

≈ 1.5 min

THE TEAM LEADER RETURNS FOR BUSINESS SUSTAINABILITY + CLOSING

PRESENTER 1

UCHEKWE UCHECHUKWU HEPHZIBAH

Team Leader & Frontend Developer

PRESENTATION TOPIC

Introduction & Frontend Development

THIS SECTION COVERS

1. Project background, aim and intended users
2. Implemented marketplace workflows
3. Frontend technologies and component structure
4. Real interface screens and user interaction

P1

Finding a tutor is a multi-factor matching problem.

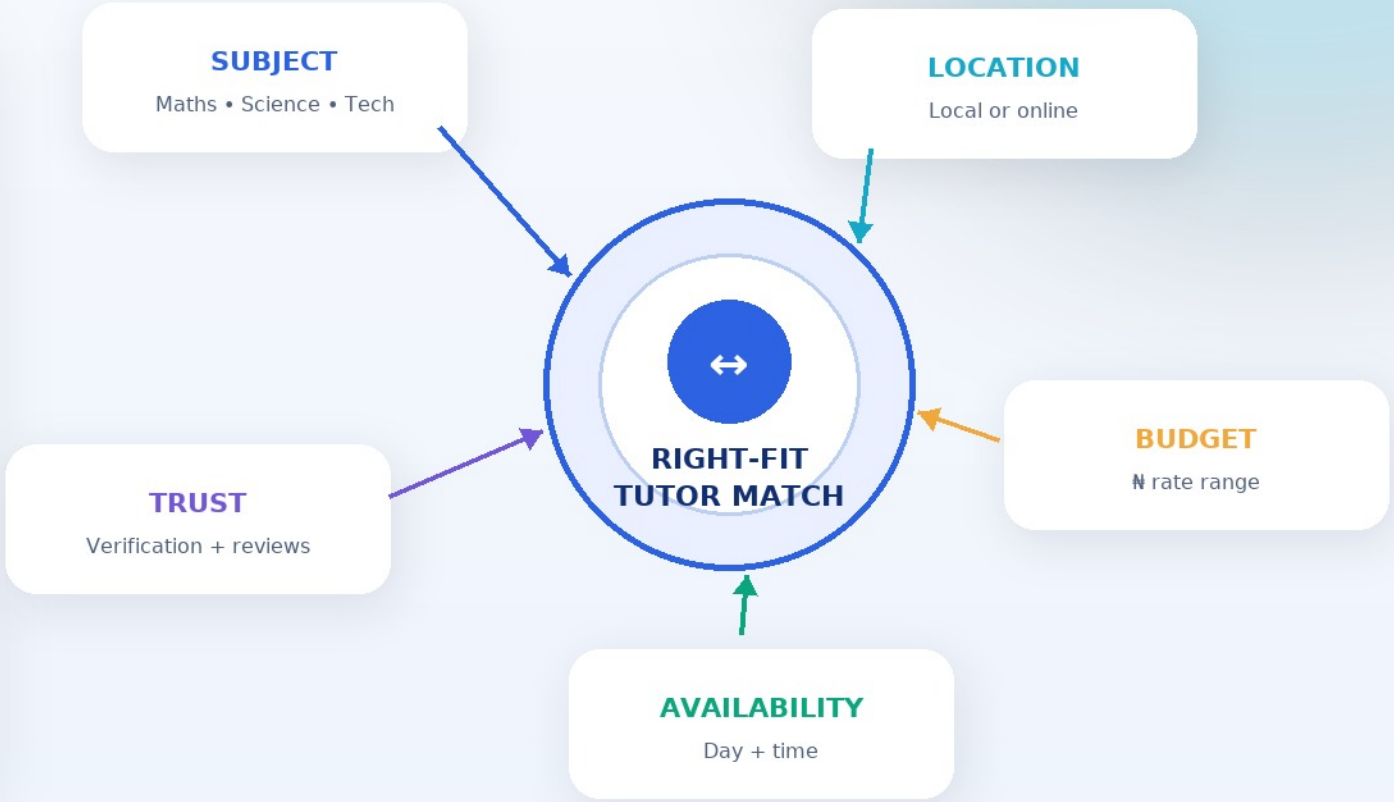
The challenge is not tutor discovery alone—it is finding the right fit with enough information to decide confidently.

PROBLEM STATEMENT

Students and parents need a way to compare suitable tutors by subject, location or online mode, budget, availability, curriculum and trust.

! Informal search channels make these factors hard to compare consistently.

NO UNSUPPORTED RESEARCH STATISTICS USED



One marketplace, four roles, one connected workflow.

Aim: help Nigerian learners discover suitable tutors and manage the journey from search to completed-session review.

1 LEARNER SIDE

STUDENT / PARENT

Search • save • book • message • review

2 SUPPLY SIDE

TUTOR

Profile • subjects • availability • requests

3 TRUST LAYER

ADMIN

Account review • verification • oversight

IMPLEMENTED USER JOURNEY

PAYMENTS ARE NOT CONFIRMED

1 **DISCOVER** Filters → 2 **EVALUATE** Profiles → 3 **REQUEST** Bookings → 4 **CONNECT** Messages → 5 **MANAGE** Dashboards → 6 **TRUST** Reviews + admin

Frontend stack + interface navigation

Typed React components, responsive styling, motion and a clear route from discovery to management.

TECHNOLOGIES

1. **NEXT.JS 16**
   App Router • pages + layouts
2. **REACT 19**
   Reusable interactive components
3. **TYPESCRIPT**
   Typed props, users, tutors, bookings
4. **TAILWIND CSS 4**
   Responsive design tokens + utilities
5. **FRAMER MOTION**
   Hero and interface transitions
6. **LUCIDE-REACT**
   Consistent accessible icon system

USER INTERFACE FLOW

Navbar • TutorCard • BookingModal • DashboardShell

01 DISCOVER
Homepage
Hero • subjects • calls to action

02 MATCH
Tutor directory
Search • filters • sorting

03 EVALUATE
Tutor profile
Credentials • reviews • calendar

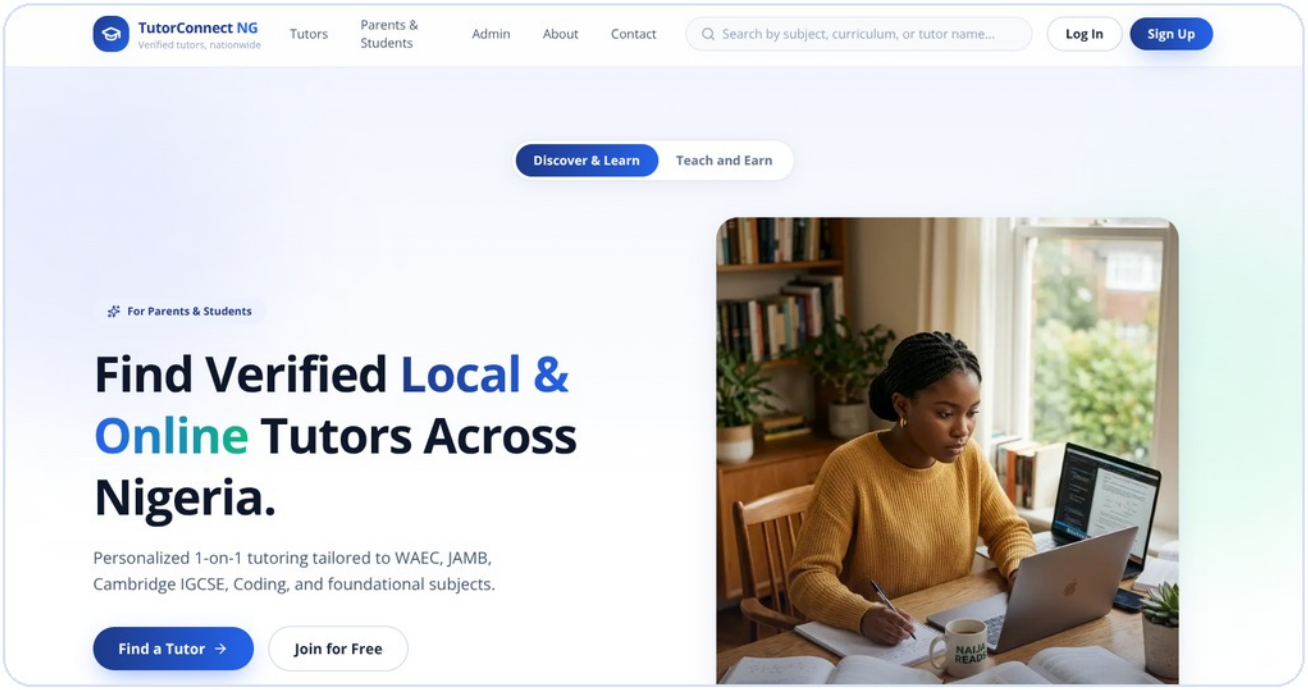
04 REQUEST
Booking modal
Subject • date • mode • estimate

05 MANAGE
Role dashboard
Status • chat • profile • review

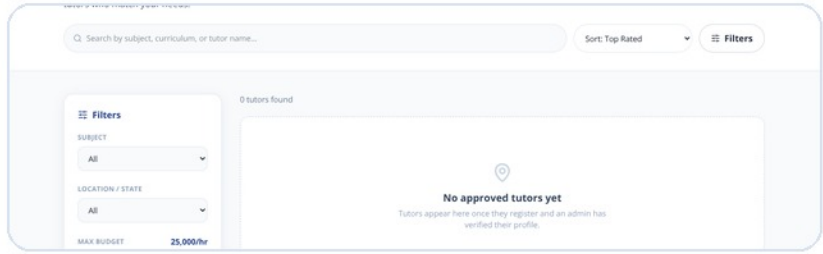
Real screens from the running application

Captured locally from this repository • no stock image is presented as an application screenshot.

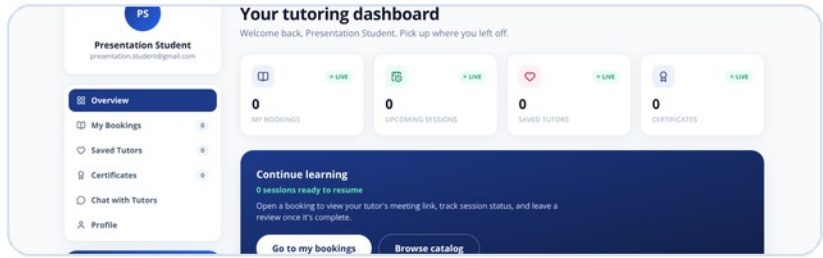
LOCAL CAPTURE: DATABASE_URL NOT CONFIGURED



01 HOMEPAGE — Discovery entry point + primary actions



02 TUTOR SEARCH — Filters + honest empty state



03 DASHBOARD — Role-based management hub

HANDOVER 01 → 02

FRONTEND COMPLETE.
THE NEXT LAYER IS THE BACKEND.





BACKEND + API

NEXT PRESENTER

CHIOMA HELEN EZEUFONNA

Backend Developer

“That concludes my presentation on frontend development and user interface design. I will now hand over to Chioma Helen Ezeufonna…”

PRESENTER 2

CHIOMA HELEN EZEUFONNA

Backend Developer

PRESENTATION TOPIC

Backend Development & API Integration

THIS SECTION COVERS

1. Purpose of the server-side layer
2. Next.js route handlers and API operations
3. bcrypt password hashing + JWT authentication
4. Authorization, validation and database requests

P2

One Next.js codebase, five API route groups.

Server-side route handlers validate requests, apply access rules and coordinate application data.

API ROUTE SURFACE

/api/auth — POST • GET • PUT • DELETE
Accounts, profiles, admin + verification

/api/tutors — GET
Verified listings + filters

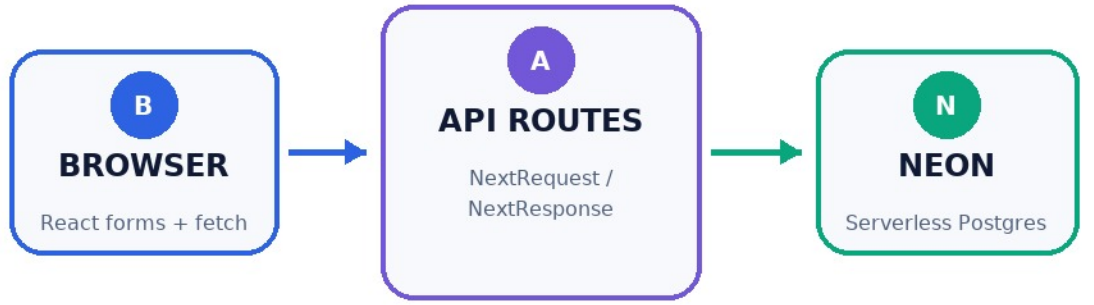
/api/bookings — GET • POST • PATCH • DELETE
Request lifecycle + ownership

/api/reviews — GET • POST
Completed-session reviews

/api/messages — GET • POST
Private one-to-one threads

REQUEST ARCHITECTURE

NODE.JS RUNTIME • JSON RESPONSES



AUTH
bcryptjs + JWT

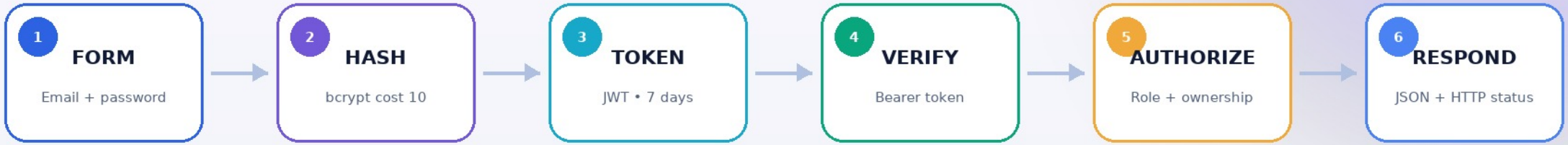
VALIDATE
JSON + required fields

AUTHORIZE
role + row ownership

MAP
SQL rows → TS objects

Authentication proves identity; authorization limits access.

A protected request moves through six deliberate checks before data returns to the interface.



AUTHENTICATION

“Who are you?”

verify signature • expiry • identity

AUTHORIZATION

“What may you do?”

role checks • ownership • scope

CURRENT TOKEN STORAGE: BROWSER localStorage

Production hardening: HttpOnly secure cookie + server middleware, while retaining API role and ownership checks.

CODE EXCERPT

```
01  const passwordHash = await bcrypt.hash(password, 10);
02  const valid = await bcrypt.compare(password, user.password_hash);
03  jwt.sign({ id, email, role }, JWT_SECRET, { expiresIn: "7d" });
04  const token = authHeader.replace("Bearer ", "");
05  const payload = verifyToken(token);
06  if (!allowedRoles.includes(user.role)) return 403;
```

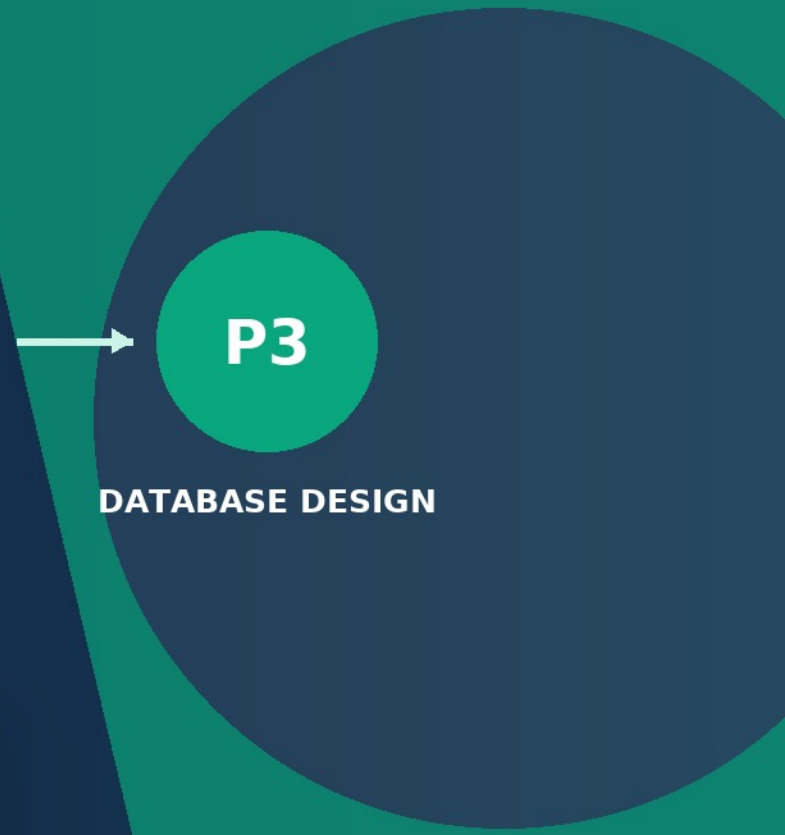
HANDOVER 02 → 03

REQUESTS ARE PROCESSED.
NOW: WHERE DOES THE DATA LIVE?

NEXT PRESENTER

UZOCHUKWU MARY CYNTHIA ENYEMAKACHUKWU

Database Developer



“I will now hand over to Uzochukwu Mary Cynthia Enyemakachukwu, who will explain how TutorConnect Nigeria stores and relates its data…”

PRESENTER 3

UZOCHUKWU MARY CYNTHIA ENYEMAKACHUKWU

Database Developer

PRESENTATION TOPIC

Database Design & Implementation

THIS SECTION COVERS

1. Purpose of persistent structured data
2. Neon Serverless PostgreSQL + SQL client
3. Current tables, constraints and triggers
4. Relationships and backend read/write flow

P3

Neon Serverless PostgreSQL: 10 tables, one source of truth.

The schema separates identity, marketplace operations and trust/communication data.

DATABASE ENGINE

NEON
SERVERLESS POSTGRESQL

@neondatabase/serverless provides a fetch-friendly SQL client suited to Vercel functions.

- Parameterized tagged SQL
- Pooled production URL

IDENTITY

users
accounts + roles

parent_profiles
child + preferences

CORE MARKETPLACE

tutor_profiles
profile + verification

tutor_subjects
subject + level

tutor_availability
weekly time slots

bookings
session requests

reviews
rating + comment

TRUST + COMMUNICATION

messages
private threads

tutor_documents
credential images

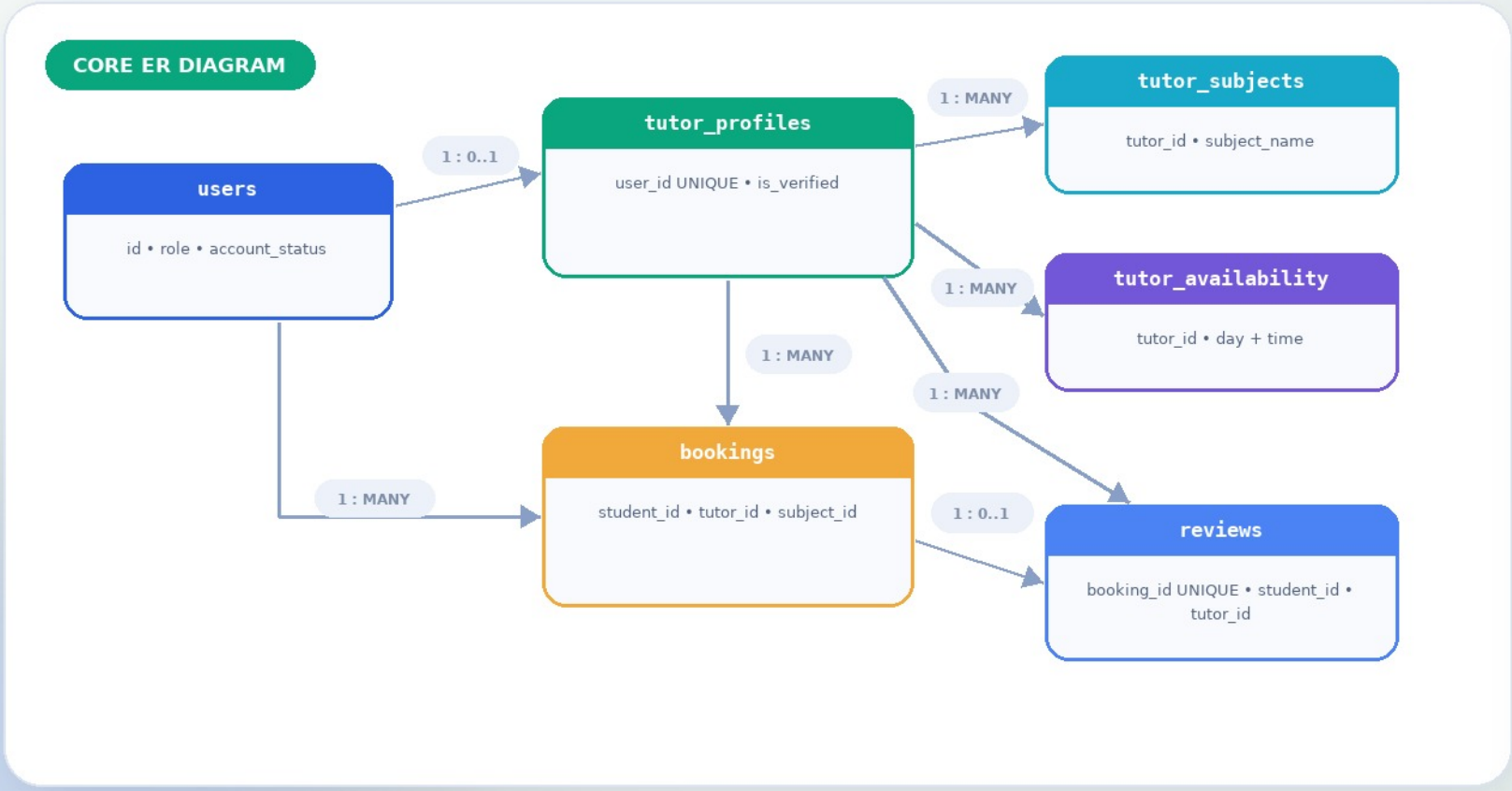
tutor_verifications
review lifecycle

INTEGRITY BY DESIGN

UNIQUE	CHECKS	FOREIGN KEYS	TRIGGERS
email • one profile/user	rating 1–5 • valid time	cascade / set null	ratings + updated_at

Core relationships connect users, tutors, bookings and review

Cardinality is based on the actual PostgreSQL foreign keys and unique constraints.



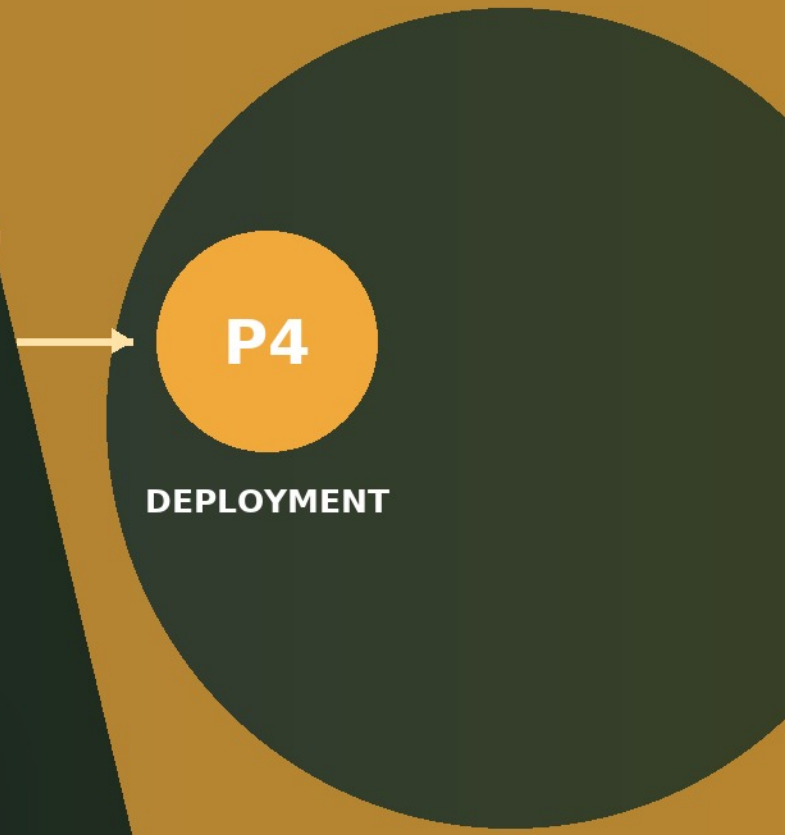
BACKEND ↔ DATABASE

1. **API ROUTE**
   validates JSON + user
2. **SQL TEMPLATE**
   parameterizes values
3. **NEON**
   reads / writes rows
4. **ROW MAPPER**
   database → TypeScript
5. **UI RESPONSE**
   JSON renders state

CASCADES CLEAN DEPENDENT ROWS

HANDOVER 03 → 04

THE DATA MODEL IS READY.
NOW: MAKE THE PRODUCT AVAILABLE.



NEXT PRESENTER

BODMAN EBUKA PRECOUS

Deployment Specialist

“I will now hand over to Bodman Ebuka Precous, our deployment specialist, who will explain how TutorConnect Nigeria is hosted and made available online.”

PRESENTER 4

BODMAN EBUKA PRECOUS

Deployment Specialist

PRESENTATION TOPIC

Deployment & Hosting

THIS SECTION COVERS

1. What deployment means for a full-stack Next.js app
2. GitHub repository → Vercel workflow
3. Environment variables + Neon connectivity
4. Verification evidence, risks and improvements

P4

GitHub → Vercel → Next.js runtime → Neon.

The repository and GitHub deployment records confirm the hosting arrangement without exposing secret values.

01 GITHUB

zibahofficial/

TutorConnectNg

Source + branch commits

02 VERCEL

Import + build

Next.js auto-detection

03 WEB + API

Pages + Node routes

One deployed application

04 NEON

Pooled Postgres

Persistent production data

RUNTIME CONFIGURATION

DATABASE_URL	Neon pooled connection
JWT_SECRET	token signing + verification
NEXT_PUBLIC_APP_URL	public application address

EVIDENCE

Repository homepage	tutor-connect-ng.vercel.app
Vercel Preview records	SUCCESS • 14 OCT 2026
vercel.json	Not required / not present
Production endpoint	Not exercised in this review

Verify what is observed. Label what is still untested.

Deployment quality comes from repeatable checks, durable storage and visible operational evidence.

OBSERVED IN REVIEW

- ✓ **Local homepage**
  loaded successfully
- ✓ **Tutor directory**
  filters + empty state loaded
- ✓ **Signup + dashboard**
  temporary student flow loaded
- ✓ **Vercel Preview records**
  successful deployments recorded
- ! **Public production URL**
  not exercised from sandbox
- ! **Local production build**
  blocked by Google Fonts access

BUILD LIMITATION WAS ENVIRONMENTAL — NOT REPORTED AS A PASS

PRODUCTION IMPROVEMENTS

01 **SELF-HOST FONTS**
Remove build-time dependency on Google Fonts.

02 **AUTOMATE CI**
Lint, build, API smoke tests and preview checks.

03 **HARDEN AUTH**
HttpOnly cookies, middleware and rate limiting.

04 **DURABLE MEDIA**
Object storage + signed access for credentials.

05 **OPERATE RELIABLY**
Monitoring, alerts, Neon backups and pooling.

06 **NO DEMO STATE**
Never rely on in-memory fallback in production.

RETURN 04 → TEAM LEADER

FOUR TECHNICAL LAYERS COVERED.
NOW: SUSTAINABILITY + CONCLUSION.

FRONTEND | BACKEND | DATABASE | DEPLOYMENT



TEAM LEADER RETURNS

RETURNING PRESENTER

UCHEKWE UCHECHUKWU HEPHZIBAH

Proposed Revenue Model • Future Sustainability • Conclusion

“I will now hand the presentation back to our team leader, Uchekwe Uchechukwu Hephzibah, who will discuss the proposed revenue model, future sustainability, and our conclusion.”

Proposed revenue model & business sustainability

Future opportunity only—not current functionality.

PROPOSED • NOT IMPLEMENTED

01 COMMISSION

Completed sessions

An agreed percentage of future in-platform payments.

02 OPTIONAL PLANS

Tutor subscriptions

Premium tools only when they provide clear extra value.

03 FEATURED LISTINGS

Labelled promotion

Paid visibility with clear labels; verification still governs trust.

ILLUSTRATION ONLY

₦10,000 × **10%** = **₦1,000**

session fee | example rate | platform gross before expenses

LAUNCH REQUIREMENTS

- Secure payment integration
- Transparent fees + refunds
- Tutor adoption + customer support
- Fraud, tax + compliance controls
- Operating-cost discipline

PRESENTER 1 • CLOSING

TUTORCONNECT NIGERIA

A credible foundation for tutor discovery, trust, booking and relationship management.

01 **CLEAR EXPERIENCE**
Search, profiles, booking and dashboards

02 **CONNECTED BACKEND**
API routes, JWT and ownership checks

03 **STRUCTURED DATA**
Neon PostgreSQL relationships + integrity

04 **DEPLOYABLE PATH**
GitHub, Vercel and operational next steps





?

QUESTIONS?

NEXT PRIORITIES

Harden authentication • self-host fonts • automate tests • verify production end to end • add secure payments before monetisation

THANK YOU